注意：

允許學生個人、非營利性的圖書館或公立學校合理使用本基金會網站所提供之書籍、各項試題及其解答。**可直接下載而不須申請**。

重版、系統地複製或大量重製這些資料的任何部分，必須獲得財團法人臺北市九章數學教育基金會的授權許可。

申請此項授權請電郵 chiuchang.tw@gmail.com

Notice:

Individual students, nonprofit libraries, or schools are permitted to make fair use of the books, papers and its solutions. Republication, systematic copying, or multiple reproduction of any part of this material is permitted only under license from the Chiuchang Mathematics Foundation.

Requests for such permission should be made by e-mailing Mr. Wen-Hsien SUN chiuchang.tw@gmail.com

International Mathematics Assessments for Schools (2025-2026)

Upper Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the **answer sheet**. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the **answer sheet** to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Upper Primary Division Round 2

Questions 1 to 5, 4 marks each

1. A gallon is defined as the volume of a rectangular prism with dimensions 11 inches × 7 inches × 3 inches. A barrel is equal to 42 gallons. Let n be a positive integer such that the volume of a cube with side length n inches is closest to the volume of one barrel. What is the value of n?

（A）19　（B）20　（C）21　（D）22　（E）23

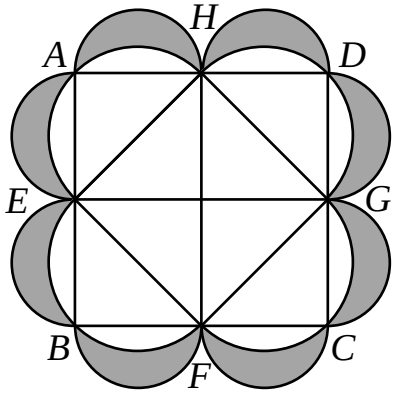
---

2. A nine-digit integer is formed by using nine different non-zero digits. Reading the digits consecutively produces 7 three-digit numbers. For example, the nine-digit number 542687931 gives us the following 7 three-digit numbers: 542, 426, 268, 687, 879, 793 and 931. Among all possible nine-digit integers formed this way, what is the minimum possible value of the sum of the seven 3-digit numbers produced?

（A）3122　（B）3132　（C）3192　（D）3202　（E）4526

---

3. As shown in the diagram below, $ABCD$ is a square, where points E, F, G and H are the midpoints of sides AB, BC, CD and DA, respectively. Semicircles are drawn with diameters EF, FG, GH, HE, AE, EB, BF, FC, CG, GD, DH and HA. If the area of square $ABCD$ is 314 cm^2, what is the total area, in cm^2, of the shaded regions? (Take $\pi = 3.14$)



（A）40　（B）$78\frac{1}{2}$　（C）80　（D）144　（E）157

---

4. In a certain country, a machine accepts only two types of coins: \$3 coins and \$5 coins, in local dollars (\$). To pay exactly \$2026, let m be the minimum possible number of coins required and let M be the maximum possible number of coins required. What is the value of $M - m$?

（A）267　（B）268　（C）269　（D）270　（E）271

---

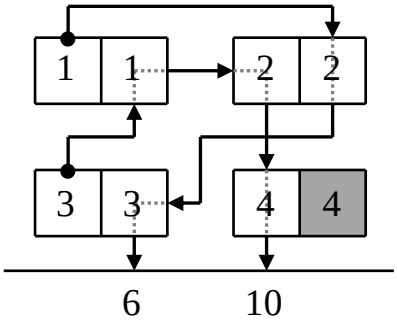
5. A factory produces two types of robots, type A robot and type B robot. In one month, the factory produced a total of 50 robots. The average weight of type A robots in kg is an integer which is equal to the number of type A robots produce. The average weight of the type B robots in kg also an integer equal to the number of type B robots produced. It is known that the factory produced fewer type A robots than type B robots, and the average weight of all 50 robots was 26 kg. How many type A robots did the factory produce that month?

（A）20　　（B）22　　（C）24　　（D）25　　（E）30

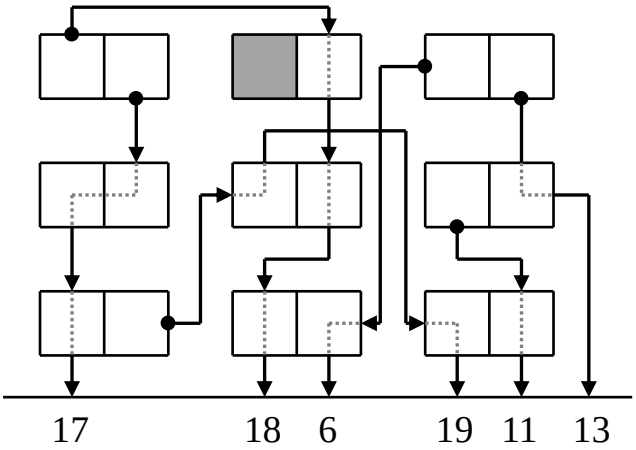
Questions 6 to 13, 5 marks each

6. What is the last non-zero digit of $1\times2\times3\times4\times...\times29\times30$?

7. In the configuration shown in the diagram below, the numbers 11, 22, 33 and 44 are placed into the boxes. There are two paths which start at the points marked by the dots and continue through the boxes, as indicated by arrows and dotted segments. The sum of the digits in the boxes along each path is shown at the bottom of the diagram. Note that the shaded box is the only box that is not on either path.



Now, in the configuration shown in the diagram below, the numbers 11, 22, 33, 44, 55, 66, 77, 88 and 99 are to be placed in the boxes. Six paths are drawn using the same rules as before, so that the shaded box is the only box not lying on any of the paths. What is the sum of all the different possible digits that can be placed in the shaded box?



8. One day, Father Bear, Mother Bear and Baby Bear ate honey. Initially, Father Bear had 120 spoonfuls of honey, Mother Bear had 80 and Baby Bear had 40.

- If only Father Bear had eaten from Mother Bear's bowl and he had eaten one third of his actual total consumption, then Baby Bear would have eaten half of the amount that Mother Bear would have had left.
- If only Father Bear had eaten from his own bowl, and he had eaten his actual total consumption, then Mother Bear would have eaten half of the amount of honey that Father Bear would have had left.
- If only Mother Bear had eaten from Baby Bear's bowl, and she had eaten one third of her actual total consumption, then Father Bear would have eaten the amount that Baby Bear would have had left.

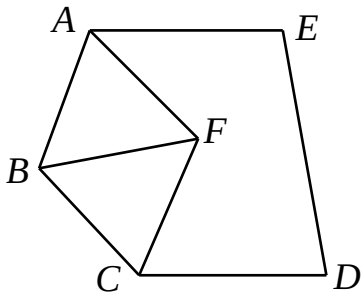
How many spoonfuls of honey did all three bears eat in total?

---

9. On a robot planet, there are three types of colour-shifting robots: red, green and blue. Whenever two robots of different colours meet, both instantly transform into robots of the third colour. If two robots of the same colour meet, nothing happens. Initially, there are 68 red robots, 44 blue robots and 75 green robots. What is the minimum number of meetings required for all the robots to become the same colour?

---

10. As shown in the diagram below, $ABCDE$ is a convex pentagon such that $AB = BC$, $\angle EAB = 110°$, $\angle CDE = 80°$ and $\angle DEA = 100°$. If point F is the intersection of the bisectors of $\angle EAB$ and $\angle BCD$, what is the angle measure, in degrees, of $\angle ABF$?



---

11. A field is covered with fully grown grass. The cows on the field eat the grass at a constant rate (each cow eats the same amount per day), while the grass grows at a different constant rate. If there are 12 cows on the field, it would take 21 days for all the grass to be completely eaten, and if there are 15 cows, it would take 14 days. How many cows are needed so that all the grass on the field is completely eaten in 9 days? (Note that the amount of grass grown per day will stay constant across days, regardless of how much grass is left.)

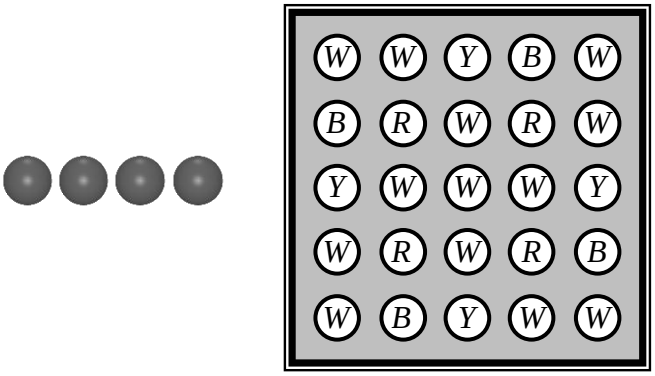
12. In the *Pawukon* calendar, the *Dwiwara* system classify each day as either a *Menga* day or a *Pepet* day based on the sum of two "ritual values", x and y, as follows:

- The value of x follows a 7-day cycle: 5, 4, 3, 7, 8, 6, 9
- The value of y follows a 5-day cycle: 9, 7, 4, 8, 5
- On a given day, if $x+y$ is even, then it is a *Menga* day;
  if $x+y$ is odd, then it is a *Pepet* day.
- On January 1, 2026, the values of x and y are $x=8$ and $y=4$.

Let a be the maximum possible number of consecutive *Menga* days and b be the maximum possible number of consecutive *Pepet* days. What is the value of $100a+b$?

---

13. In the carnival game *Wiffle Ball Toss*, a player tosses 4 balls, one at a time, onto a table with holes of four different colours: white, blue, yellow and red.



Points are scored depending on the colour of the hole each ball lands in, as shown below:

Colour	**White**	**Blue**	**Yellow**	**Red**
Score	0 points	1 point	2 points	4 points

If the total score from the four tosses is 4 or more points, the player wins a prize. For example, the sequence Yellow-Blue-White-Blue results in a total score of $2+1+0+1=4$ points, so the player gets a prize. While the sequence White-Blue-Blue-White scores only $0+1+1+0=2$ points, so no prize is won. Assume that each toss lands in exactly one hole and there are at least 4 holes of each colour.

How many different colour sequences result in winning a prize?

ANSWER SHEET of Questions 1 to 13

Questions 1 to 5, 4 marks each

- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct.

1.		**2.**		**3.**		**4.**		**5.**	

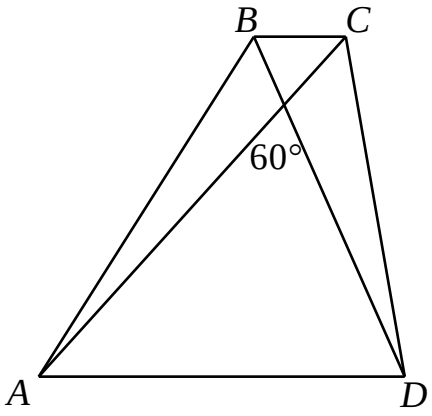
Questions 6 to 13, 5 marks each

- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited.

6.		**10.**	○
7.		**11.**	cows
8.	spoonfuls	**12.**	
9.		**13.**	colour sequences

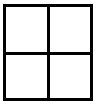
Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. As shown in the diagram below, *ABCD* is a quadrilateral, where *AD* is parallel to *BC* and $AD = 20$ cm, $BC = 6$ cm, $AC = 26$ cm. If the angle between *AC* and *BD* is $60°$, what is the difference, in cm, between the lengths of *AB* and *CD*?

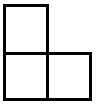


Answer : _____ cm

15. As shown in the diagram below, David has unlimited copies of the O-tetromino (left) and the V-tromino (right).

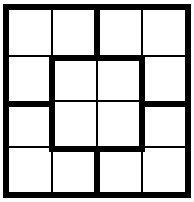


O-tetromino

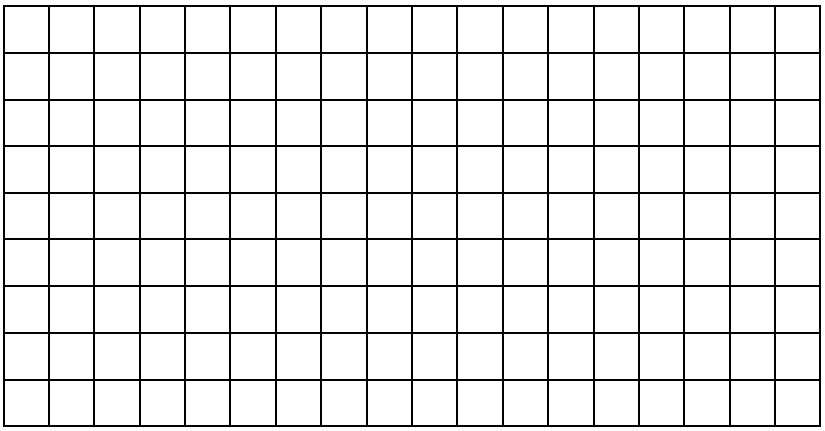


V-tromino

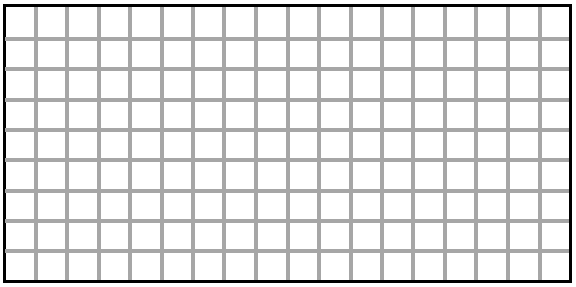
David wants to place some O-tetrominoes and V-trominoes on a game board along the grid lines to completely cover the board such that no part of any piece remains outside the board or overlapping another piece. For example, if the game board has dimensions 4×4, if David places O-tetrominoes and V-trominoes as shown in the diagram below, then the number of V-trominoes used is 4.



If the game board has dimensions 9×18, what is the minimum number of V-trominoes David needs to completely cover the board? Please prove that it is impossible to cover the board using fewer V-trominoes and illustrate one possible placement that achieves this minimum.



Answer：_____ V-trominoes,



***